

Ubuntu Mirror Server - Setup Guide

Ubuntu Mirror Server - Complete Setup Guide

Purpose: Step-by-step instructions to set up an Ubuntu Package Mirror Server for offline OS upgrades

Target Versions: Ubuntu 16.04 → 18.04 → 20.04 → 22.04 → 24.04

Time Required: 1-2 hours setup + 12-24 hours initial sync

Last Updated: January 31, 2026

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Prerequisites Checklist

Before starting, ensure you have:

-  Ubuntu 24.04 LTS server installed
-  Two disks configured:
 - OS Disk: 100 GB
 - Data Disk: 1 TB mounted at `/var/spool/apt-mirror`
-  Root or sudo access
-  Internet connectivity (100+ Mbps recommended)
-  Static IP address configured

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1. Verify Disk Configuration {anchor:verify-disk}

First, verify your disk setup is correct:

```
1 # Check disk layout
2 df -h
3
4 # Should see:
5 # /dev/sda2 or /dev/mapper/ubuntu--vg-ubuntu--lv → / (OS disk ~100 GB)
6 # /dev/sdb1 → /var/spool/apt-mirror (Data disk ~1 TB)
```

Expected output:

Filesystem	Size	Mounted On	Purpose
/dev/sda2 (or LVM)	~100 GB	/	Operating System
/dev/sdb1	~1 TB	/var/spool/apt-mirror	Mirror storage

{warning}

If `/var/spool/apt-mirror` is NOT mounted, you need to mount your data disk first. See troubleshooting section.

{warning}

2. Install Required Software {anchor:install-software}

Install `apt-mirror` and `nginx`:

```
1 # Update package lists
2 sudo apt-get update
```

```

3
4 # Install packages
5 sudo apt-get install -y apt-mirror nginx
6
7 # Verify installation
8 which apt-mirror
9 which nginx
10
11 # Check nginx is running
12 sudo systemctl status nginx

```

Expected output:

```

1 apt-mirror: /usr/bin/apt-mirror
2 nginx: /usr/sbin/nginx
3 nginx: active (running)

```

3. Configure apt-mirror {anchor:configure-apt-mirror}

Create Configuration File

```

1 # Edit the apt-mirror configuration
2 sudo nano /etc/apt/mirror.list

```

Configuration for FULL Mirror (660 GB)

Delete all existing content and paste this:

```

1 ##### config #####
2 set base_path /var/spool/apt-mirror
3 set mirror_path $base_path/mirror
4 set skel_path $base_path/skel
5 set var_path $base_path/var
6 set cleanscript $var_path/clean.sh
7 set defaultarch amd64
8 set nthreads 20
9 set _tilde 0
10 ##### end config #####
11
12 # Ubuntu 16.04 LTS (Xenial)
13 deb http://archive.ubuntu.com/ubuntu xenial main restricted universe
14 multiverse
15 deb http://archive.ubuntu.com/ubuntu xenial-updates main restricted
16 universe multiverse
17 deb http://archive.ubuntu.com/ubuntu xenial-security main restricted
18 universe multiverse
19
20 # Ubuntu 18.04 LTS (Bionic)
21 deb http://archive.ubuntu.com/ubuntu bionic main restricted universe
22 multiverse
23 deb http://archive.ubuntu.com/ubuntu bionic-updates main restricted
24 universe multiverse
25 deb http://archive.ubuntu.com/ubuntu bionic-security main restricted
26 universe multiverse
27

```

```

22 # Ubuntu 20.04 LTS (Focal)
23 deb http://archive.ubuntu.com/ubuntu focal main restricted universe
   multiverse
24 deb http://archive.ubuntu.com/ubuntu focal-updates main restricted
   universe multiverse
25 deb http://archive.ubuntu.com/ubuntu focal-security main restricted
   universe multiverse
26
27 # Ubuntu 22.04 LTS (Jammy)
28 deb http://archive.ubuntu.com/ubuntu jammy main restricted universe
   multiverse
29 deb http://archive.ubuntu.com/ubuntu jammy-updates main restricted
   universe multiverse
30 deb http://archive.ubuntu.com/ubuntu jammy-security main restricted
   universe multiverse
31
32 # Ubuntu 24.04 LTS (Noble)
33 deb http://archive.ubuntu.com/ubuntu noble main restricted universe
   multiverse
34 deb http://archive.ubuntu.com/ubuntu noble-updates main restricted
   universe multiverse
35 deb http://archive.ubuntu.com/ubuntu noble-security main restricted
   universe multiverse
36
37 clean http://archive.ubuntu.com/ubuntu

```

Save and exit: `Ctrl+X`, `Y`, `Enter`

{info}

For MINIMAL mirror (305 GB - main and restricted only):

Remove `universe multiverse` from all `deb` lines above.

{info}

Verify Configuration

```

1 # Check file exists and has content
2 ls -lh /etc/apt/mirror.list
3
4 # Verify base_path is NOT commented out
5 cat /etc/apt/mirror.list | grep "set base_path"
6 # Should show: set base_path /var/spool/apt-mirror
7
8 # Create necessary directories
9 sudo mkdir -p /var/spool/apt-mirror/{mirror,skel,var}
10
11 # Set permissions
12 sudo chown -R root:root /var/spool/apt-mirror
13 sudo chmod -R 755 /var/spool/apt-mirror

```

Test Configuration

```

1 # Test apt-mirror (will start downloading)
2 sudo apt-mirror
3
4 # You should see:

```

```

5 # Downloading 570 index files using 20 threads...
6 # Begin time: ...
7
8 # Press Ctrl+C to stop (we'll do full sync later)

```

4. Configure nginx Web Server {anchor:configure-nginx}

Create nginx Site Configuration

```

1 # Create new site configuration
2 sudo nano /etc/nginx/sites-available/apt-mirror

```

Paste this configuration:

```

1 server {
2     listen 80 default_server;
3     listen [::]:80 default_server;
4
5     server_name _;
6
7     root /var/spool/apt-mirror/mirror;
8     autoindex on;
9
10    location / {
11        try_files $uri $uri/ =404;
12    }
13
14    location /ubuntu {
15        alias /var/spool/apt-mirror/mirror/archive.ubuntu.com/ubuntu;
16        autoindex on;
17    }
18
19    # Logging
20    access_log /var/log/nginx/apt-mirror-access.log;
21    error_log /var/log/nginx/apt-mirror-error.log;
22 }

```

Save and exit: Ctrl+X, Y, Enter

Enable the Site

```

1 # Enable apt-mirror site
2 sudo ln -sf /etc/nginx/sites-available/apt-mirror /etc/nginx/sites-
  enabled/
3
4 # Disable default site
5 sudo rm -f /etc/nginx/sites-enabled/default
6
7 # Test nginx configuration
8 sudo nginx -t
9 # Should show: nginx: configuration file /etc/nginx/nginx.conf test is
  successful
10
11 # Restart nginx
12 sudo systemctl restart nginx

```

```
13 sudo systemctl enable nginx
14
15 # Verify nginx is running
16 sudo systemctl status nginx
```

5. Set Up Automated Daily Sync {anchor:setup-automation}

Create systemd Service

```
1 # Create service file
2 sudo nano /etc/systemd/system/apt-mirror.service
```

Paste:

```
1 [Unit]
2 Description=APT Mirror Sync Service
3 After=network-online.target
4
5 [Service]
6 Type=oneshot
7 ExecStart=/usr/bin/apt-mirror
8 StandardOutput=append:/var/log/apt-mirror.log
9 StandardError=append:/var/log/apt-mirror.log
10 User=root
```

Save and exit: Ctrl+X , Y , Enter

Create systemd Timer

```
1 # Create timer file
2 sudo nano /etc/systemd/system/apt-mirror.timer
```

Paste:

```
1 [Unit]
2 Description=Daily APT Mirror Sync
3
4 [Timer]
5 OnCalendar=daily
6 OnCalendar=02:00
7 Persistent=true
8
9 [Install]
10 WantedBy=timers.target
```

Save and exit: Ctrl+X , Y , Enter

Enable Timer (Don't Start Yet)

```
1 # Reload systemd
2 sudo systemctl daemon-reload
3
```

```

4 # Enable timer (will start after first manual sync)
5 sudo systemctl enable apt-mirror.timer
6
7 # Verify timer is configured
8 systemctl list-timers apt-mirror.timer

```

6. Create Monitoring Scripts {anchor:monitoring-scripts}

Create Health Check Script

```

1 # Create monitoring script
2 sudo nano /usr/local/bin/mirror-status.sh

```

Paste this complete script:

```

1 #!/bin/bash
2 # mirror-status.sh - Mirror Server Health Check
3
4 echo "
5 echo "    Ubuntu Mirror Server - Health Check    "
6 echo "
7 echo ""
8
9 # Mirror disk space
10 echo "📊 Mirror Disk Usage:"
11 df -h /var/spool/apt-mirror | tail -1
12 USAGE=$(df /var/spool/apt-mirror | tail -1 | awk '{print $5}' | sed
13 's/%//')
14 if [ "$USAGE" -gt 90 ]; then
15     echo "⚠️ WARNING: Disk usage above 90%"
16 elif [ "$USAGE" -gt 80 ]; then
17     echo "⚠️ CAUTION: Disk usage above 80%"
18 fi
19 echo ""
20
21 # Mirror size
22 echo "📦 Current Mirror Size:"
23 MIRROR_SIZE=$(du -sh /var/spool/apt-mirror/mirror/ 2>/dev/null | cut -
24 f1)
25 if [ -n "$MIRROR_SIZE" ]; then
26     echo "$MIRROR_SIZE"
27 else
28     echo "Mirror sync in progress or not started..."
29 fi
30 echo ""
31
32 # Sync status
33 echo "🔄 Sync Process:"
34 if pgrep -f "apt-mirror" > /dev/null; then
35     echo "✅ Sync is currently RUNNING"
36     PACKAGES=$(find /var/spool/apt-mirror/mirror -name "*.deb"
37 2>/dev/null | wc -l)
38     echo "    Downloaded packages: $PACKAGES"
39 else
40     echo "🛑 No sync currently running"
41     if [ -f /var/log/apt-mirror.log ]; then

```

```

39     LAST_SYNC=$(stat -c %y /var/log/apt-mirror.log | cut -d' ' -
f1,2 | cut -d'.' -f1)
40     echo "    Last sync: $LAST_SYNC"
41     fi
42 fi
43 echo ""
44
45 # Nginx status
46 echo "🌐 Nginx Status:"
47 if systemctl is-active --quiet nginx; then
48     echo "✅ Nginx is running"
49 else
50     echo "❌ Nginx is not running"
51 fi
52 echo ""
53
54 # Ubuntu versions
55 echo "📁 Available Ubuntu Versions:"
56 for version in xenial bionic focal jammy noble; do
57     if [ -d "/var/spool/apt-
mirror/mirror/archive.ubuntu.com/ubuntu/dists/$version" ]; then
58         SIZE=$(du -sh "/var/spool/apt-
mirror/mirror/archive.ubuntu.com/ubuntu/dists/$version" 2>/dev/null |
cut -f1)
59         echo "    ✅ $version ($SIZE)"
60     else
61         echo "    ⌚ $version (not yet synced)"
62     fi
63 done
64 echo ""
65
66 # HTTP access test
67 echo "🔗 HTTP Access Test:"
68 if curl -s -f http://localhost/ubuntu/dists/noble/Release > /dev/null
2>&1; then
69     echo "✅ Local HTTP access working"
70 else
71     echo "⌚ Local HTTP access not ready (sync may not be complete)"
72 fi
73 echo ""
74
75 echo "┌───────────────────────────────────────────────────────────────────────────────────┐"

```

Save and exit: Ctrl+X, Y, Enter

```

1 # Make executable
2 sudo chmod +x /usr/local/bin/mirror-status.sh
3
4 # Test the script
5 sudo /usr/local/bin/mirror-status.sh

```

7. Start Initial Mirror Sync {anchor:initial-sync}

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⚠ Important: Initial Sync

This step will download **~660 GB** of data and take **12-24 hours** depending on your internet speed.

Time estimates:

- 100 Mbps: 18-24 hours
 - 500 Mbps: 3-4 hours
 - 1 Gbps: 1.5-2 hours
- {panel}

Start the Sync

```

1 # Start sync in background with logging
2 sudo nohup apt-mirror > /var/log/apt-mirror-initial.log 2>&1 &
3
4 # Note the process ID
5 echo "Mirror sync started in background"
6
7 # Verify it's running
8 ps aux | grep apt-mirror | grep -v grep
9
10 # View initial output
11 tail -f /var/log/apt-mirror-initial.log

```

Expected output:

```

1 Downloading 570 index files using 20 threads...
2 Begin time: Sat Jan 31 02:XX:XX 2026
3 [20]... [19]... [18]... [17]... [16]...
4 Processing metadata files from archive.ubuntu.com...
5 Downloading packages...

```

Press **Ctrl+C** to exit log viewer (sync continues in background).

8. Monitor Sync Progress {anchor:monitor-progress}**Quick Status Commands**

```

1 # Run health check
2 sudo /usr/local/bin/mirror-status.sh
3
4 # Check if sync is running
5 ps aux | grep apt-mirror | grep -v grep
6
7 # Current mirror size
8 du -sh /var/spool/apt-mirror/mirror/
9
10 # Disk usage
11 df -h /var/spool/apt-mirror

```

```

12
13 # Count downloaded packages
14 find /var/spool/apt-mirror/mirror -name "*.deb" 2>/dev/null | wc -l
15
16 # Watch live log
17 tail -f /var/log/apt-mirror-initial.log

```

Progress Indicators

Time Elapsed	Expected Size	Status
0-10 min	~1 GB	Downloading metadata
2 hours	~100 GB	15% complete
6 hours	~300 GB	45% complete
12 hours	~500 GB	75% complete
18-24 hours	~660 GB	Complete

Monitor with watch Command

```

1 # Auto-refresh every 5 minutes
2 watch -n 300 'du -sh /var/spool/apt-mirror/mirror/ && df -h
   /var/spool/apt-mirror'

```

9. Post-Sync Configuration {anchor:post-sync}

After Sync Completes

When you see **"End time:"** in the log file:

```

1 # 1. Verify completion
2 sudo /usr/local/bin/mirror-status.sh
3
4 # 2. Run cleanup script (removes superseded packages)
5 sudo /var/spool/apt-mirror/var/clean.sh
6
7 # 3. Check final size
8 du -sh /var/spool/apt-mirror/mirror/
9 # Should show ~660 GB
10
11 # 4. Start automated daily sync timer
12 sudo systemctl start apt-mirror.timer
13
14 # 5. Verify timer is active
15 systemctl list-timers apt-mirror.timer
16 # Should show next run at 02:00 daily
17

```

```

18 # 6. Test local HTTP access
19 curl -I http://localhost/ubuntu/dists/noble/Release
20 # Should return: HTTP/1.1 200 OK

```

Get Server IP Address

```

1 # Find your mirror server IP
2 ip addr show | grep "inet " | grep -v 127.0.0.1
3
4 # Or
5 hostname -I

```

Note this IP address - you'll need it for client configuration.

10. Test from Client {anchor:test-client}

On a Test Data Processor or Ubuntu VM

```

1 # Backup current sources
2 sudo cp /etc/apt/sources.list /etc/apt/sources.list.backup
3
4 # Set mirror IP (replace with your actual IP)
5 MIRROR_IP="10.34.200.20"
6
7 # Point to mirror server
8 sudo sed -i "s|http://archive.ubuntu.com|http://$MIRROR_IP|g"
   /etc/apt/sources.list
9
10 # Update package lists
11 sudo apt-get update
12
13 # Verify mirror is being used
14 apt-cache policy | grep $MIRROR_IP

```

Expected output:

```

1 500 http://10.34.200.20/ubuntu noble/main amd64 Packages
2   release v=24.04,o=Ubuntu,a=noble...

```

Create Client Test Script

```

1 # Create connectivity test script on CLIENT
2 sudo nano /usr/local/bin/test-mirror-connectivity.sh

```

Paste:

```

1 #!/bin/bash
2 # test-mirror-connectivity.sh - Test mirror connectivity from client
3
4 MIRROR_IP="${1:-}"
5
6 if [ -z "$MIRROR_IP" ]; then
7     echo "Usage: $0 <mirror-ip>"

```

```

8     exit 1
9 fi
10
11 echo "Testing connection to mirror: $MIRROR_IP"
12 echo ""
13
14 # 1. Ping test
15 echo "1 ICMP Ping Test:"
16 if ping -c 3 -W 5 "$MIRROR_IP" > /dev/null 2>&1; then
17     echo "    ✓ Can ping $MIRROR_IP"
18 else
19     echo "    ⚠ Cannot ping (may be blocked by firewall)"
20 fi
21 echo ""
22
23 # 2. TCP port 80
24 echo "2 TCP Port 80 Test:"
25 if timeout 5 bash -c "cat < /dev/null > /dev/tcp/$MIRROR_IP/80"
26 2>/dev/null; then
27     echo "    ✓ Port 80 is reachable"
28 else
29     echo "    ✗ Port 80 NOT reachable"
30 fi
31 echo ""
32
33 # 3. HTTP GET
34 echo "3 HTTP GET Test:"
35 HTTP_CODE=$(curl -s -o /dev/null -w "%{http_code}" --max-time 10
36 "http://$MIRROR_IP/ubuntu/dists/noble/Release")
37 if [ "$HTTP_CODE" = "200" ]; then
38     echo "    ✓ HTTP GET successful (code: $HTTP_CODE)"
39 else
40     echo "    ✗ HTTP failed (code: $HTTP_CODE)"
41 fi
42 echo ""
43
44 # 4. Package availability
45 echo "4 Package Availability Test:"
46 AVAILABLE=0
47 for version in xenial bionic focal jammy noble; do
48     if curl -s -f --max-time 10
49     "http://$MIRROR_IP/ubuntu/dists/$version/Release" > /dev/null 2>&1;
50     then
51         echo "    ✓ $version available"
52         ((AVAILABLE++))
53     else
54         echo "    ✗ $version NOT available"
55     fi
56 done
57 echo ""
58 if [ $AVAILABLE -eq 5 ]; then
59     echo "    ✓ SUCCESS: All Ubuntu versions available - mirror is
60 ready!"
61     exit 0
62 else
63     echo "    ⚠ WARNING: Only $AVAILABLE/5 versions available"
64     exit 1
65 fi

```

63 [fi](#)**Save and make executable:**

```

1 sudo chmod +x /usr/local/bin/test-mirror-connectivity.sh
2
3 # Run test
4 sudo /usr/local/bin/test-mirror-connectivity.sh 10.34.200.20

```

11. Troubleshooting {anchor:troubleshooting}**Issue:** apt-mirror shows "invalid config file"**Symptoms:**

```

1 apt-mirror: invalid config file specified at /usr/bin/apt-mirror line
  138.

```

Solution:

```

1 # Check if config file exists
2 ls -la /etc/apt/mirror.list
3
4 # Verify base_path is NOT commented out
5 cat /etc/apt/mirror.list | grep "^set base_path"
6 # Should NOT have # at the beginning
7
8 # If commented, edit file and remove # from these lines:
9 # set base_path      /var/spool/apt-mirror
10 # set mirror_path    $base_path/mirror
11 # set skel_path       $base_path/skel
12 # set var_path        $base_path/var
13 # set cleanscript     $var_path/clean.sh

```

Issue: /var/spool/apt-mirror not mounted**Symptoms:**

```

1 df -h
2 # /var/spool/apt-mirror is not shown

```

Solution:

```

1 # Check if data disk exists
2 lsblk
3 # Look for 1TB disk (probably /dev/sdb)
4
5 # If disk exists but not mounted:
6 sudo mkdir -p /var/spool/apt-mirror
7 sudo mount /dev/sdb1 /var/spool/apt-mirror
8

```

```

9 # Add to /etc/fstab for persistence
10 UUID=$(sudo blkid -s UUID -o value /dev/sdb1)
11 echo "UUID=$UUID /var/spool/apt-mirror ext4 defaults,noatime 0 2" |
  sudo tee -a /etc/fstab
12
13 # Verify
14 df -h /var/spool/apt-mirror

```

Issue: Sync is very slow

Symptoms:

- Download speed < 10 MB/s
- Sync taking longer than expected

Solution:

```

1 # 1. Check internet speed
2 curl -s https://raw.githubusercontent.com/sivel/speedtest-
  cli/master/speedtest.py | python3 -
3
4 # 2. Verify nthreads setting
5 grep nthreads /etc/apt/mirror.list
6 # Should be: set nthreads      20
7
8 # 3. Check if other processes are using bandwidth
9 sudo iftop
10
11 # 4. Consider using faster mirror (not archive.ubuntu.com)
12 # Edit /etc/apt/mirror.list and replace with regional mirror

```

Issue: nginx not serving files

Symptoms:

```

1 curl http://localhost/ubuntu/dists/noble/Release
2 # Returns 404 or connection refused

```

Solution:

```

1 # Check nginx is running
2 sudo systemctl status nginx
3
4 # Check nginx configuration
5 sudo nginx -t
6
7 # Check file permissions
8 ls -la /var/spool/apt-mirror/mirror/
9
10 # Check nginx error log
11 sudo tail -50 /var/log/nginx/error.log

```

```
12
13 # Restart nginx
14 sudo systemctl restart nginx
```

Issue: Disk space full during sync

Symptoms:

```
1 | No space left on device
```

Solution:

```
1 # Check disk usage
2 df -h /var/spool/apt-mirror
3
4 # If full, you may need to:
5 # Option 1: Use minimal mirror (remove universe/multiverse)
6 # Option 2: Expand disk (if cloud/VM)
7 # Option 3: Remove unnecessary Ubuntu versions
8
9 # For cloud expansion (AWS/OCI):
10 # 1. Expand volume in cloud console (1TB → 2TB)
11 # 2. Extend partition:
12 sudo growpart /dev/sdb 1
13 sudo resize2fs /dev/sdb1
14
15 # Verify new size
16 df -h /var/spool/apt-mirror
```

Issue: Sync stopped/crashed

Symptoms:

- Process not running
- Log shows errors

Solution:

```
1 # Check if process is running
2 ps aux | grep apt-mirror
3
4 # Check last log entries
5 tail -100 /var/log/apt-mirror-initial.log
6
7 # Restart sync
8 sudo nohup apt-mirror > /var/log/apt-mirror-restart.log 2>&1 &
9
10 # apt-mirror will resume from where it stopped
```

Maintenance Commands

Daily Operations

```

1 # Check mirror status
2 sudo /usr/local/bin/mirror-status.sh
3
4 # Check if daily sync ran
5 systemctl list-timers apt-mirror.timer
6
7 # View sync logs
8 sudo tail -50 /var/log/apt-mirror.log
9
10 # Check disk space
11 df -h /var/spool/apt-mirror

```

Weekly Maintenance

```

1 # Run cleanup (removes superseded packages)
2 sudo /var/spool/apt-mirror/var/clean.sh
3
4 # Check for errors
5 sudo journalctl -u apt-mirror.service | grep -i error
6
7 # Verify nginx access logs
8 sudo tail -100 /var/log/nginx/apt-mirror-access.log

```

Quick Reference

Important Files and Directories

Path	Purpose
/etc/apt/mirror.list	apt-mirror configuration
/var/spool/apt-mirror/mirror/	Downloaded packages
/var/log/apt-mirror.log	Sync logs
/etc/nginx/sites-available/apt-mirror	nginx configuration
/usr/local/bin/mirror-status.sh	Health check script

Important Commands

Command	Purpose
<code>sudo apt-mirror</code>	Start manual sync
<code>sudo /usr/local/bin/mirror-status.sh</code>	Check mirror health
<code>sudo systemctl status nginx</code>	Check web server
<code>systemctl list-timers apt-mirror.timer</code>	Check automated sync schedule
<code>df -h /var/spool/apt-mirror</code>	Check disk space

Success Checklist

After completing setup, verify:

- ☒ Mirror sync completed successfully (~660 GB)
- ☒ All 5 Ubuntu versions available (xenial, bionic, focal, jammy, noble)
- ☒ nginx serving files on port 80
- ☒ Daily sync timer enabled and scheduled
- ☒ Health check script working
- ☒ Test client can access mirror
- ☒ HTTP access test passes

Next Steps

After mirror server is operational:

1. **Update Confluence functional spec** with actual mirror IP
2. **Configure firewall rules** to allow client access (port 80)
3. **Test with pilot Data Processors** before production

4. **Document client configuration** procedures
 5. **Set up monitoring/alerting** for disk space and sync failures
 6. **Create runbook** for support team
-

Related Documentation

- Ubuntu Mirror Server - Functional Spec v2.0
 - Ubuntu Mirror Server - DevOps Guide
 - Client Configuration Procedures
 - Troubleshooting Guide
-

Document Status:  Complete

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